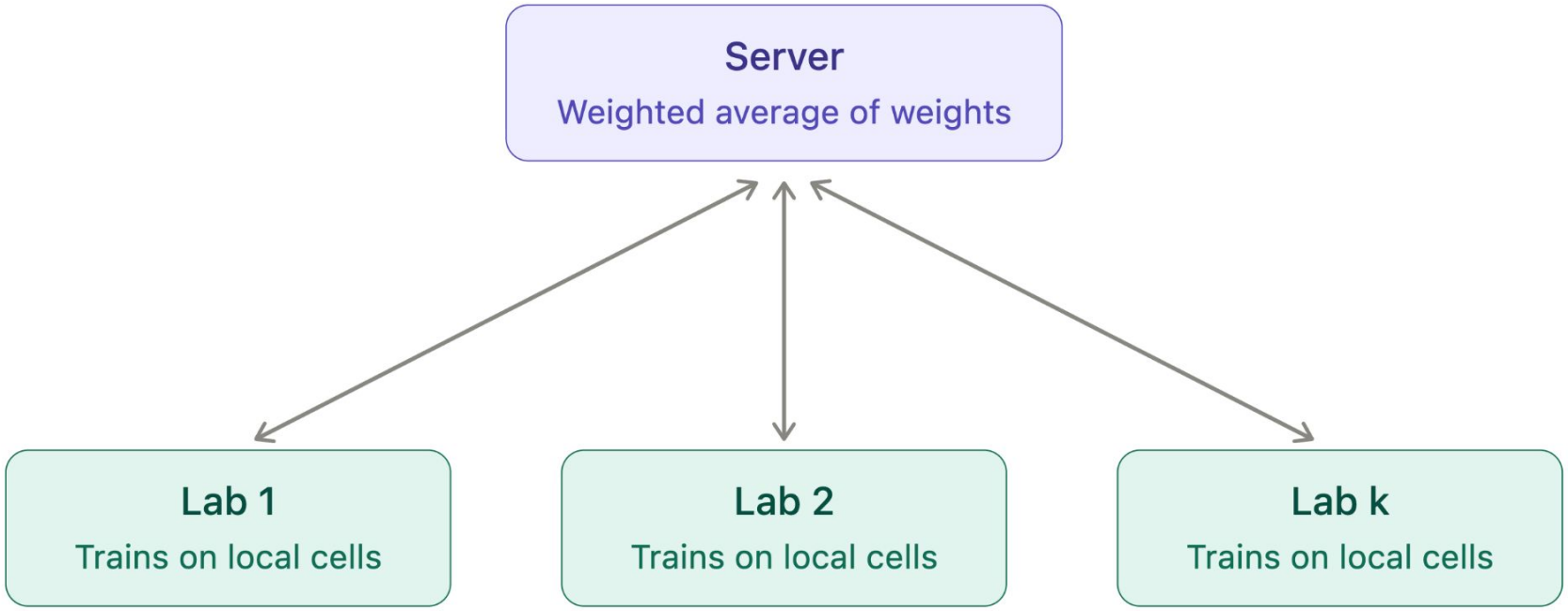
Federated Bureau of Investigation

Annika Laberge, Ernest Glukhov, Kamen Todorov, Matthew Goldman, Zahra Zare, Mukund Varma

What is Federated Learning?



weights and sample counts only. Data never leave the lab.

Why now?

- Biological data inherently carry privacy concerns (DNA / Image)
- The data exist but sit in silos
- The tooling is mature. Flower, NVIDIA FLARE, and [15 other] frameworks were compared on FAIR criteria in 2026 (Chavero-Diez et al.). Flower in particular has a user-friendly interface and good documentation

Team

Infrastructure and security team

- Annika Laberge
- Ernest Glukhov
- Kamen Todorov

Data engineering and modeling

- Matthew Goldman
- Zahra Zare

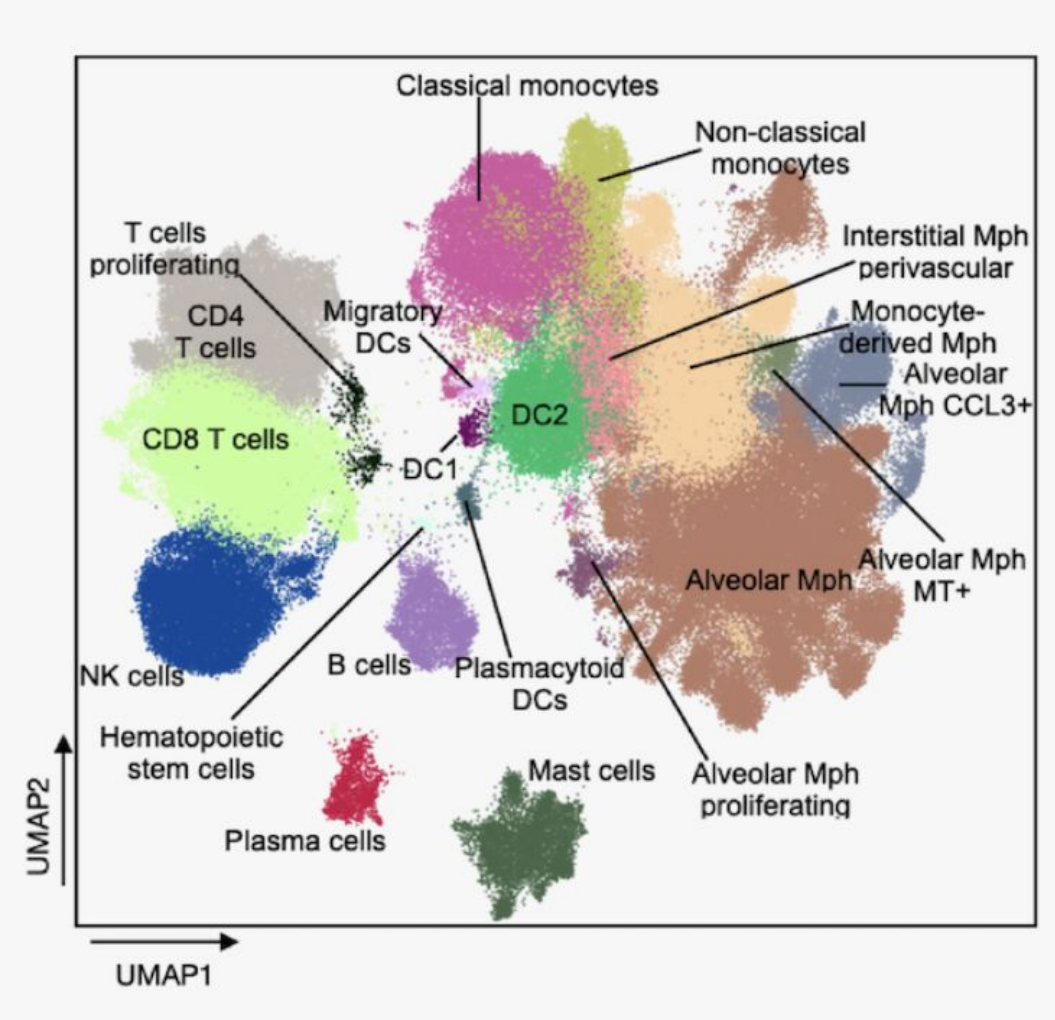
Team Lead

- Mukund Varma

Cell-type classification on Human Lung Cell Atlas

What if the different labs had not been able to share the primary data with each other or with a central training lab?

If we can’t bring the data to the model, maybe we can take the model to the data!



Data preprocessing

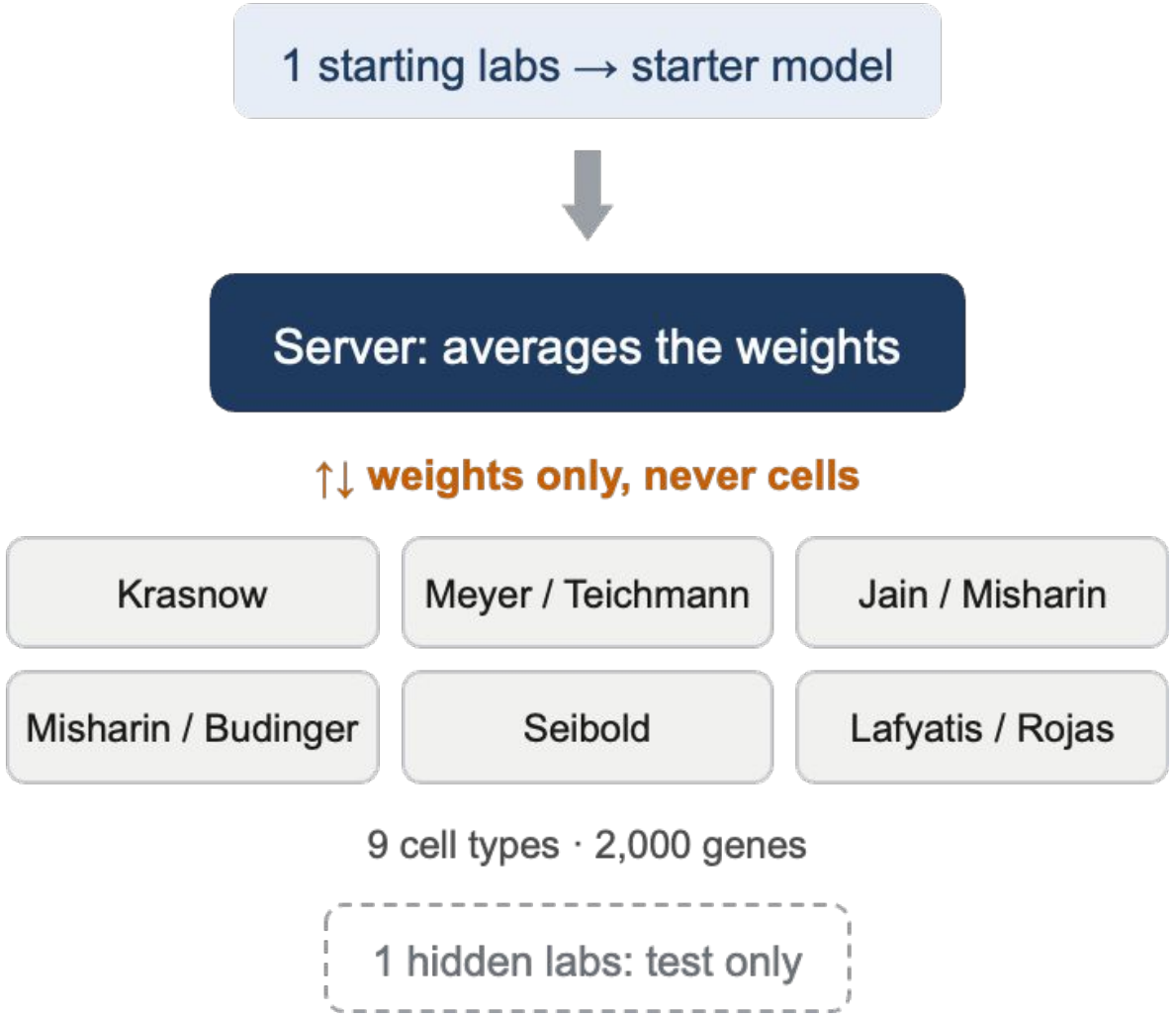
- Each lab / laptop / client has unique datasets that made up the HLCA
- Subset to a single set of 2000 HVGs that will be sent to each client so every client has the same set of features

How the HLCA core was split

Group	Source datasets	Cells	Donors	Cell types
Initial training	Banovich_Kropski_2020, Barbry_Leroy_2020	196,203	48	9
Evaluation	Nawijn_2021, Misharin_2021	135,243	12	9
Client 0	Krasnow_2020	60,944	3	9
Client 1	Meyer_2019, Teichmann_Meyer_2019	47,720	9	8
Client 2	Jain_Misharin_2021_10Xv1, Jain_Misharin_2021_10Xv2	45,557	7	6
Client 3	Misharin_Budinger_2018	41,216	8	8
Client 4	Seibold_2020_10Xv2, Seibold_2020_10Xv3	33,593	14	7
Client 5	Lafyatis_Rojas_2019_10Xv1, Lafyatis_Rojas_2019_10Xv2	24,178	6	8
Total	**14 datasets**	**584,654**	**107**	**9**

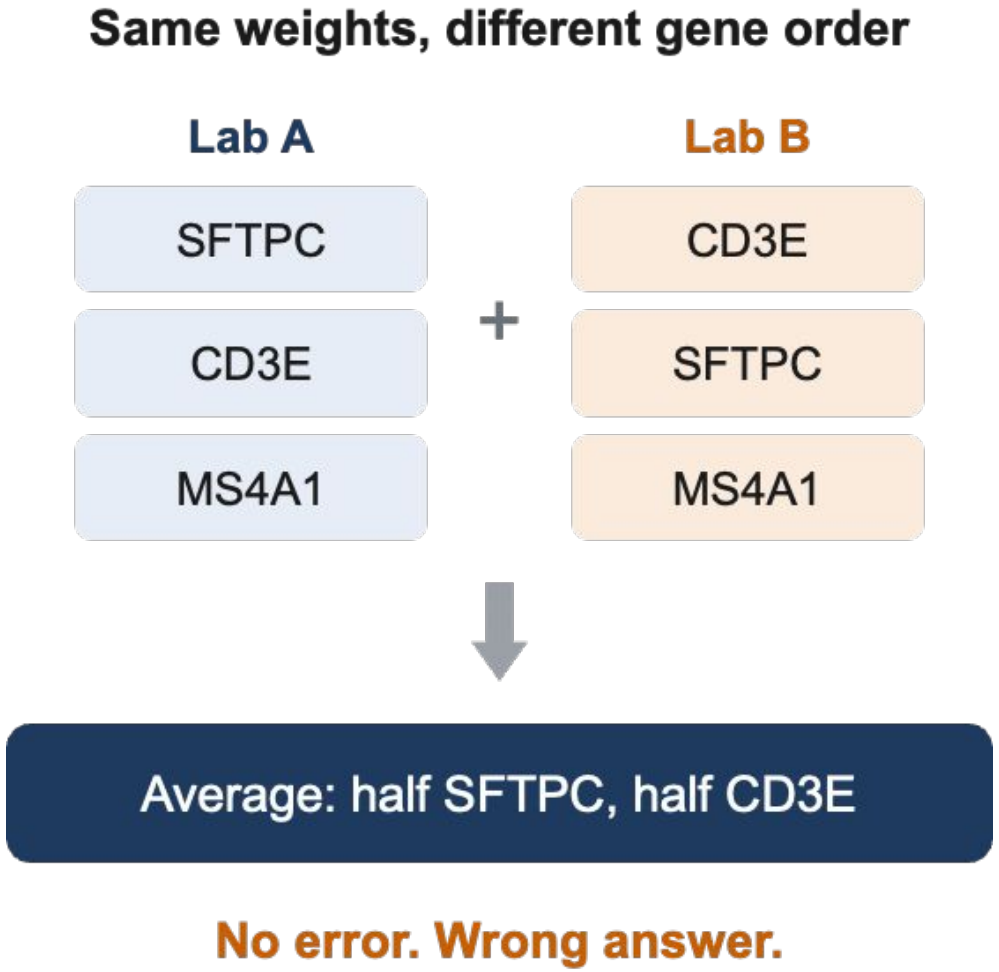
Six labs, one model, no shared cells

- Splitting the atlas by the lab
- Only model weights leave each lab
- Test on a hidden lab



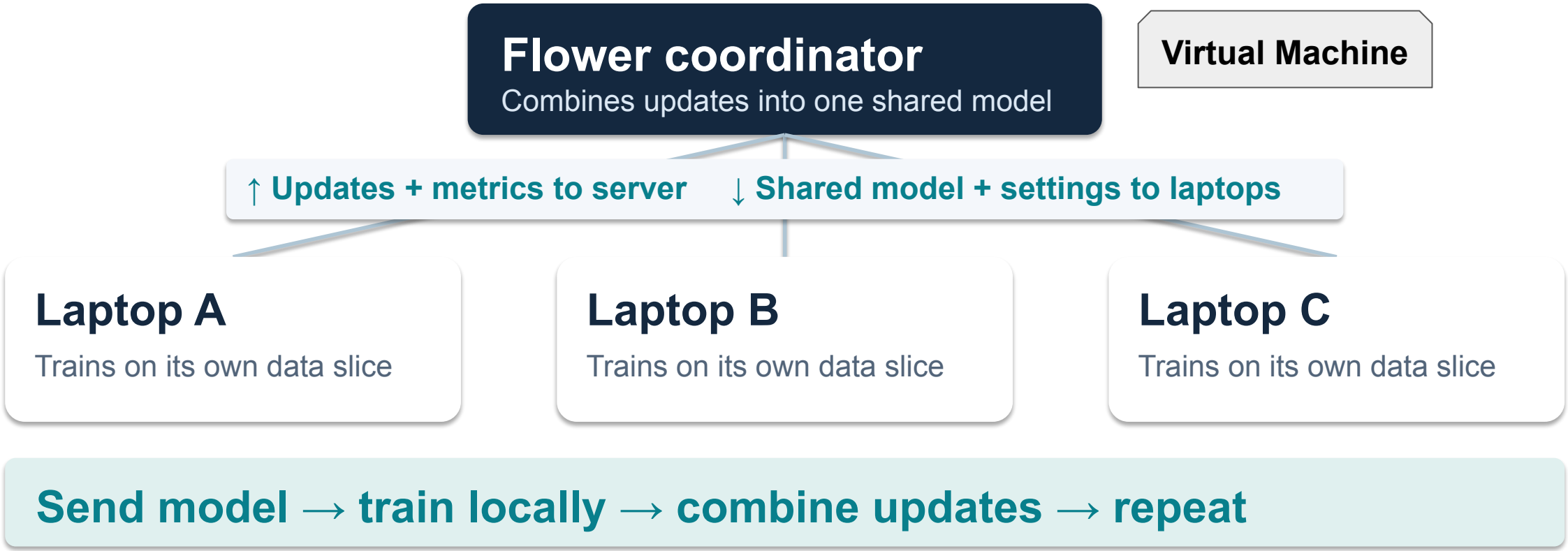
Why this is harder than it sounds?

- Every lab is different
- mistakes are silent



Architecture: based on Flower [python]

Training individually on each laptop. Weights sent back to the coordinator to update shared model



Demo

Client readiness status

Misconfiguration diagnosis

Training and results visualization

From a trusted demo to a controlled deployment

Keeping raw data local is a start—not a complete security or privacy guarantee.

01 Control who connects

Verify servers and approve each node.

Separate participant and administrator access.

Protect the dashboard; revoke lost-device access.

02 Control what runs

Run only site-approved training code.

Isolate it from personal files and credentials.

Limit permissions, network access and resources.

03 Protect shared results

Evaluate secure aggregation and privacy limits.

Review models, metrics and gene attributions.

Require approval before using sensitive data.

04 Prepare for failures

Validate updates and keep an audit trail.

Protect stored models, logs and backups.

Test access revocation and recovery.

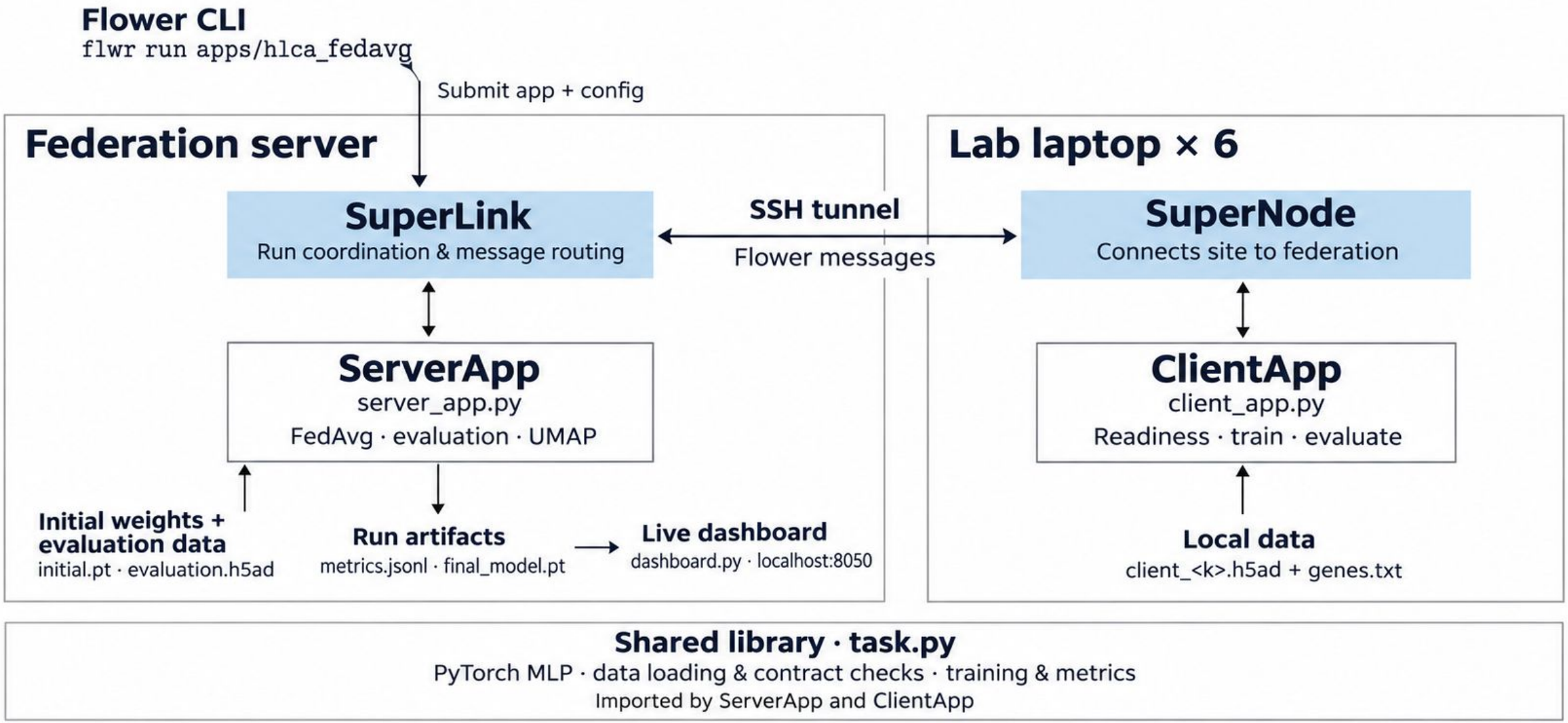
Next: harden and test on public data before considering a sensitive-data pilot.

Future recommendations—not implemented protections or a security certification. | Source: SECURITY_ROADMAP.md

Backup

Software architecture

HLCA federated learning · Flower + PyTorch



Server → sites: global weights + config

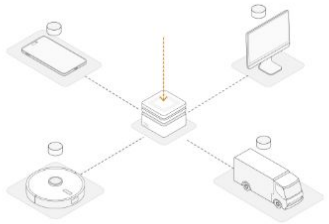
Sites → server: local weights, counts + sampled embeddings

Solution: Federated training on HLCA Using Flower Framework

Federated learning in five steps

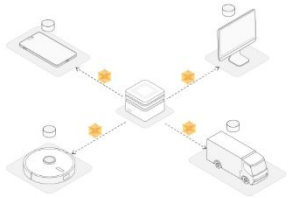
Step 0: Initialize global model

We start by initializing the model on the server. This is exactly the same in classic centralized learning: we initialize the model parameters, either randomly or from a previously saved checkpoint.



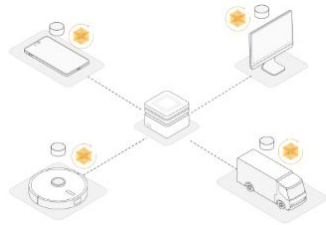
Step 1: Send model to a number of connected organizations/devices (client nodes)

Next, we send the parameters of the global model to the connected client nodes (think: edge devices like smartphones or servers belonging to organizations). This is to ensure that each participating node starts its local training using the same model parameters. We often use only a few of the connected nodes instead of all nodes. The reason for this is that selecting more and more client nodes has diminishing returns.



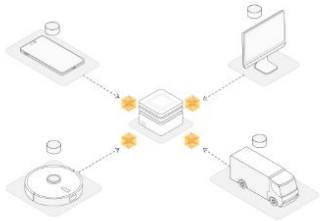
Step 2: Train model locally on the data of each organization/device (client node)

Now that all (selected) client nodes have the latest version of the global model parameters, they start the local training. They use their own local dataset to train their own local model. They don't train the model until full convergence, but they only train for a little while. This could be as little as one epoch on the local data, or even just a few steps (mini-batches).



Step 3: Return model updates back to the server

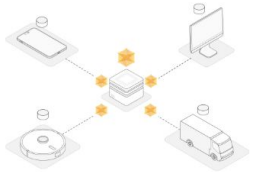
After local training, each client node has a slightly different version of the model parameters they originally received. The parameters are all different because each client node has different examples in its local dataset. The client nodes then send those model updates back to the server. The model updates they send can either be the full model parameters or just the gradients that were accumulated during local training.



Step 4: Aggregate model updates into a new global model

The server receives model updates from the selected client nodes. If it selected 100 client nodes, it now has 100 slightly different versions of the original global model, each trained on the local data of one client. But didn't we want to have one model that contains the learnings from the data of all 100 client nodes?

In order to get one single model, we have to combine all the model updates we received from the client nodes. This process is called *aggregation*, and there are many different ways to do it. The most basic way is called *Federated Averaging* (McMahan et al., 2016), often abbreviated as *FedAvg*. *FedAvg* takes the 100 model updates and, as the name suggests, averages them. To be more precise, it takes the *weighted average* of the model updates, weighted by the number of examples each client used for training. The weighting is important to make sure that each data example has the same "influence" on the resulting global model. If one client has 10 examples, and another client has 100 examples, then - without weighting - each of the 10 examples would influence the global model ten times as much as each of the 100 examples.



Step 5: Repeat steps 1 to 4 until the model converges

Steps 1 to 4 are what we call a single round of federated learning. The global model parameters get sent to the participating client nodes (step 1), the client nodes train on their local data (step 2), they send their updated models to the server (step 3), and the server then aggregates the model updates to get a new version of the global model (step 4).

During a single round, each client node that participates in that iteration only trains for a little while. This means that after the aggregation step (step 4), we have a model that has been trained on all the data of all participating client nodes, but only for a little while. We then have to repeat this training process over and over again to eventually arrive at a fully trained model that performs well across the data of all client nodes.

Join the team. Start a run. Watch it learn.

The lead starts the server; laptops connect before training begins. Everyone can follow progress on the dashboard.

1 Prepare

Use the demo branch.

Install with make env.

Set up SSH key login.

2 Connect & launch

Lead starts server services.

Laptops run the join script.

Lead starts training.

3 Watch

Open localhost:8050.

Follow accuracy and loss.

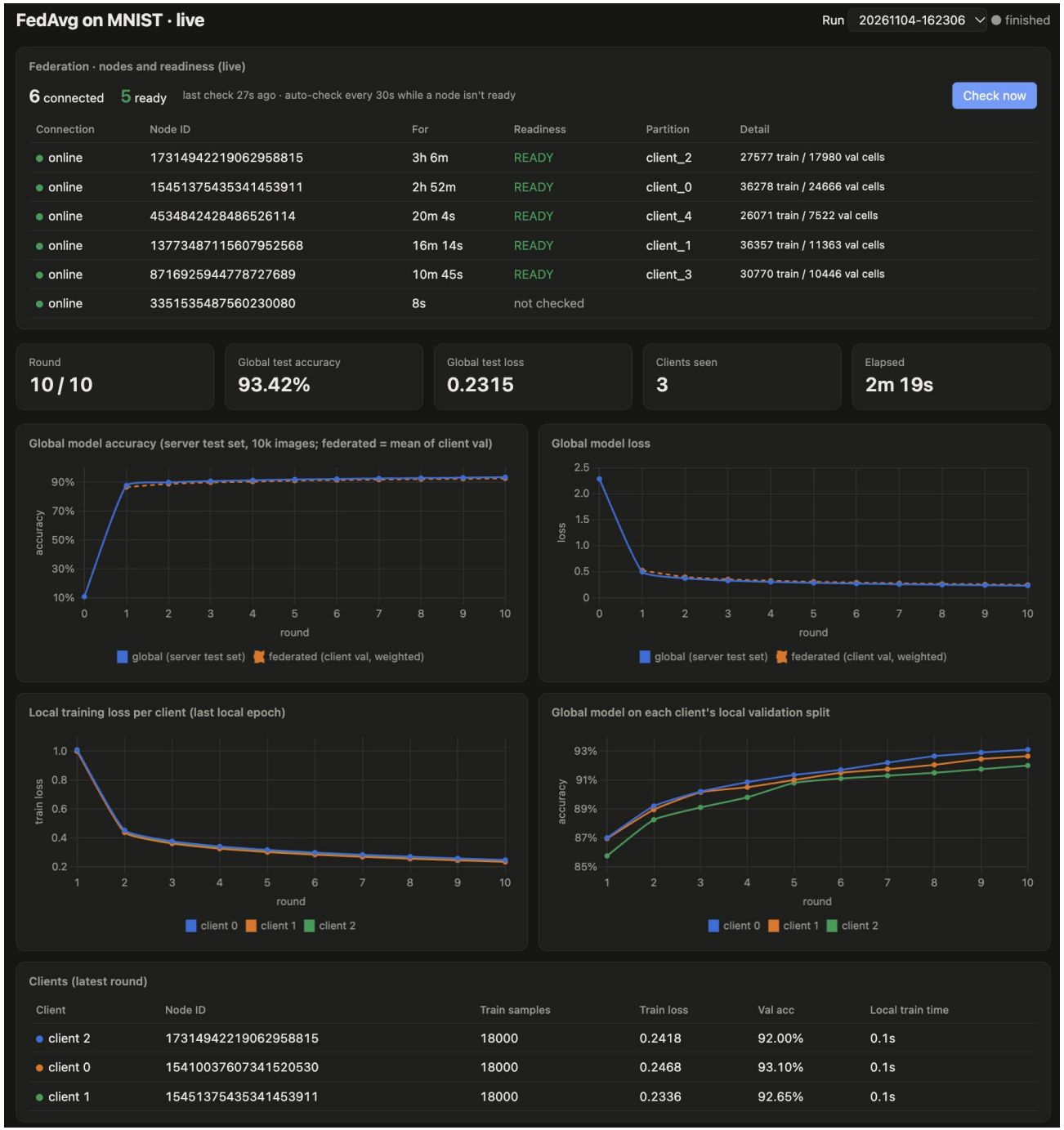
Ctrl+C stops your node.

Raw training images stay local. SSH protects the connection.

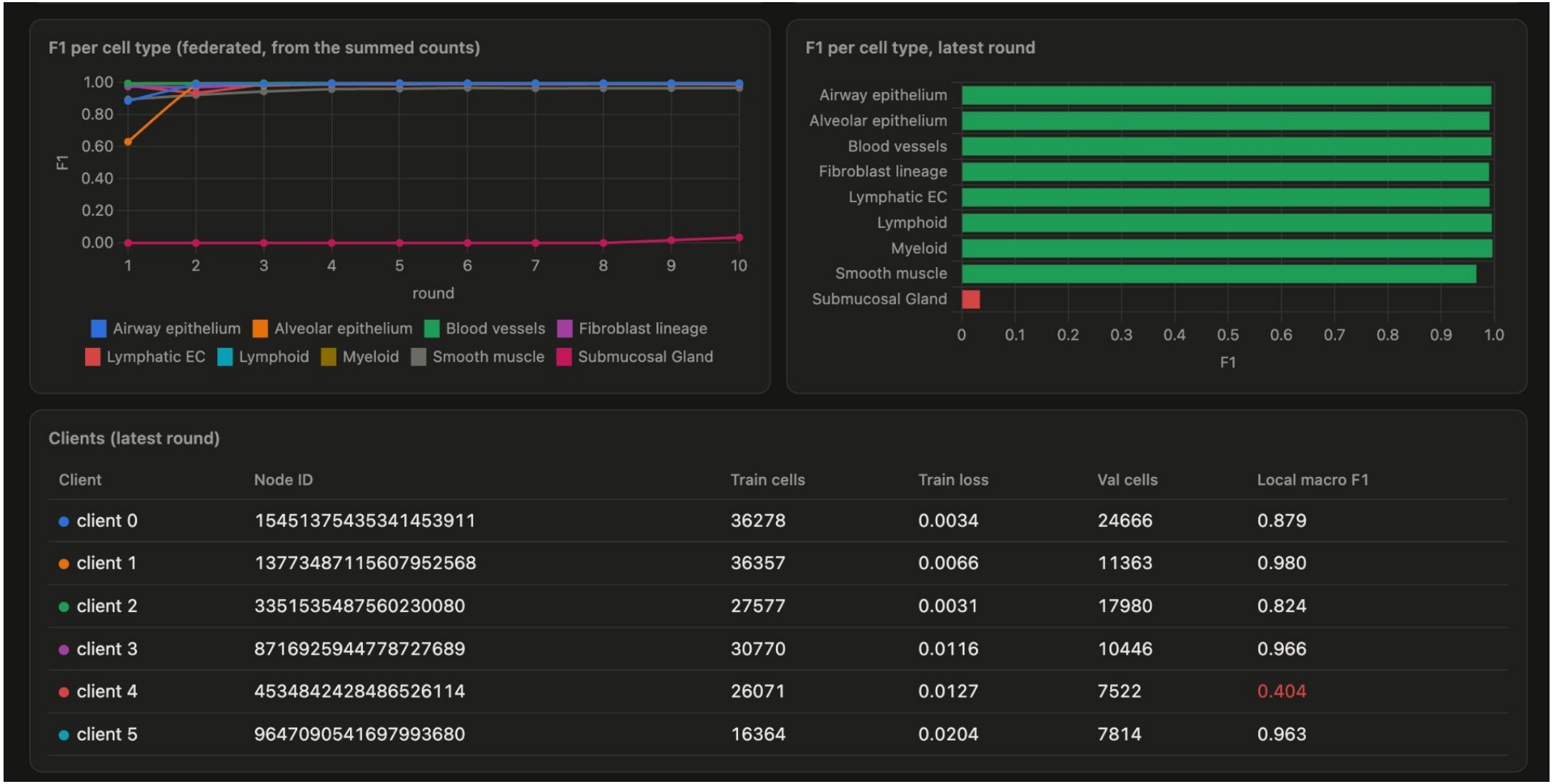
Demo security only: no differential privacy or secure aggregation.

A 120-second response deadline skips missing replies—not the laptop permanently.

Monitoring


FedAvg on MNIST · live
Run 20261104-162306
finished
Federation · nodes and readiness (live)
6 connected 5 ready last check 27s ago · auto-check every 30s while a node isn't ready
Check now
Connection Node ID For Readiness Partition Detail
online 17314942219062958815 3h 6m READY client_2 27577 train / 17980 val cells
online 15451375435341453911 2h 52m READY client_0 36278 train / 24666 val cells
online 4534842428486526114 20m 4s READY client_4 26071 train / 7522 val cells
online 13773487115607952568 16m 14s READY client_1 36357 train / 11363 val cells
online 8716925944778727689 10m 45s READY client_3 30770 train / 10446 val cells
online 3351535487560230080 8s not checked
Round
10 / 10
Global test accuracy
93.42%
Global test loss
0.2315
Clients seen
3
Elapsed
2m 19s
Global model accuracy (server test set, 10k images; federated = mean of client val)
accuracy
round
global (server test set)
federated (client val, weighted)
Global model loss
loss
round
global (server test set)
federated (client val, weighted)
Local training loss per client (last local epoch)
train loss
round
client 0
client 1
client 2
Global model on each client's local validation split
accuracy
round
client 0
client 1
client 2
Clients (latest round)
Client Node ID Train samples Train loss Val acc Local train time
client 2 17314942219062958815 18000 0.2418 92.00% 0.1s
client 0 15410037607341520530 18000 0.2468 93.10% 0.1s
client 1 15451375435341453911 18000 0.2336 92.65% 0.1s

Pitfalls: what happens if only one split holds a cell-type

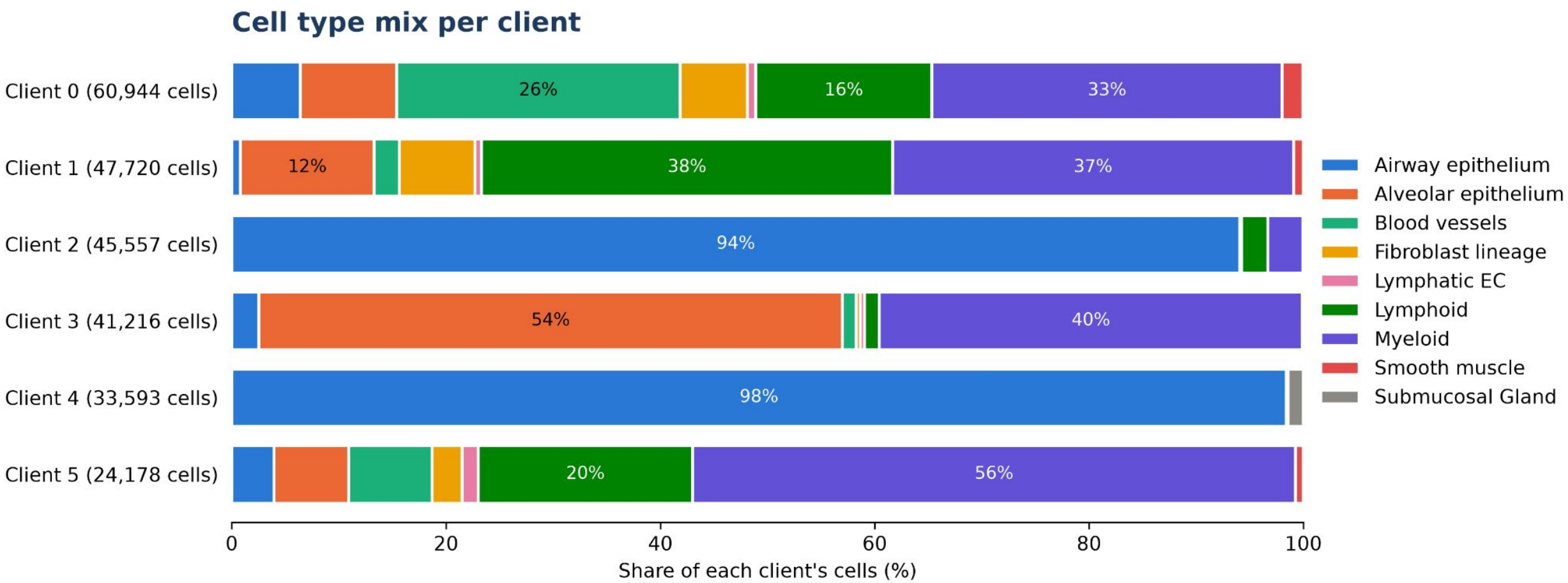


Clients (latest round)

Client	Node ID	Train cells	Train loss	Val cells	Local macro F1
client 0	15451375435341453911	36278	0.0034	24666	0.879
client 1	13773487115607952568	36357	0.0066	11363	0.980
client 2	3351535487560230080	27577	0.0031	17980	0.824
client 3	8716925944778727689	30770	0.0116	10446	0.966
client 4	4534842428486526114	26071	0.0127	7522	0.404
client 5	9647090541697993680	16364	0.0204	7814	0.963

Cells per cell type in each group

Cell type	Initial training	Evaluation	Client 0	Client 1	Client 2	Client 3	Client 4	Client 5	All groups
Airway epithelium	75,105	49,716	3,907	384	42,857	1,050	33,053	951	207,023
Myeloid	48,642	37,473	19,936	17,864	1,503	16,285	8	13,604	155,315
Lymphoid	18,780	20,506	10,002	18,292	1,117	564	25	4,835	74,121
Alveolar epithelium	25,682	9,106	5,460	5,959	0	22,429	14	1,692	70,342
Blood vessels	17,525	6,346	16,143	1,124	9	541	5	1,878	43,571
Fibroblast lineage	3,484	9,537	3,830	3,368	53	155	0	673	21,100
Submucosal Gland	3,530	639	28	0	18	0	485	0	4,700
Lymphatic EC	2,420	889	463	292	0	161	3	367	4,595
Smooth muscle	1,035	1,031	1,175	437	0	31	0	178	3,887
Total	**196,203**	**135,243**	**60,944**	**47,720**	**45,557**	**41,216**	**33,593**	**24,178**	**584,654**


Cell type mix per client
Client 0 (60,944 cells)
Client 1 (47,720 cells)
Client 2 (45,557 cells)
Client 3 (41,216 cells)
Client 4 (33,593 cells)
Client 5 (24,178 cells)
26%
16%
33%
12%
38%
37%
94%
54%
40%
98%
20%
56%
0
20
40
60
80
100
Share of each client's cells (%)
Airway epithelium
Alveolar epithelium
Blood vessels
Fibroblast lineage
Lymphatic EC
Lymphoid
Myeloid
Smooth muscle
Submucosal Gland